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Editorial Office

Journal of Neural Engineering

Dear Editors,

Please consider my manuscript, "Source-only model selection and limits of fixed spectral-sharing pipelines in cross-subject motor-imagery EEG decoding," for publication as an original research paper.

The study examines source-only model selection and the transfer limits of fixed spectral-sharing pipelines. BNCI development analyses are followed by separately frozen binary evaluations on PhysioNet, Cho2017 and Lee2019_MI. Zero calibration means no target-dependent fitting or model selection; label use for metadata and scoring is distinguished from learning. Results and uncertainty are reported at participant level.

The frozen Q15 evaluation includes 52 Cho2017 and 54 Lee2019_MI participants under a prespecified common 21-channel adapter. The shared-input pipeline had 1.513 percentage points lower balanced accuracy on Cho2017 (95% paired-person bootstrap interval -2.249 to -0.804); the Lee2019_MI difference was small and uncertain ($+0.204$ points, interval -0.515 to $+0.969$). These adverse and uncertain results are retained. The contrasts compare complete pipelines, including their source-selected training durations.

A separate BNCI sensor analysis uses fixed baseline/task intervals and participant-level μ/β descriptors, specified after decoder outcomes were known and before the power calculation. Its associations with saved source-LOSO scores are descriptive and do not feed back into model selection. The analysis distinguishes recorded physiology from evidence about decoder mechanisms. It includes nine BNCI participants and has no external physiological replication.

The accompanying code, saved predictions, frozen contracts and validation records make the evaluation traceable. Q15 validation reconstructs raw-data processing, frozen-checkpoint outputs and prescribed statistics without target fitting. Independent verification of the BNCI analysis includes full central-sensor periodogram replay and all saved-table aggregations, checked against its committed pre-power recipe and execution manifest. Physical export voltage calibration, the original Cho acquisition reference and hardware cue latency remain unresolved.

The contribution is an empirical audit of source-only model selection and the transfer limits of fixed spectral-sharing pipelines, rather than a new decoding algorithm. Person-level development gains, frozen external contrasts and separate sensor-level descriptions show why an aggregate improvement need not imply broad benefit or a physiological decoder mechanism. These questions concern the reliability and interpretation of BCI evidence, fitting the scope of Journal of Neural Engineering. The study does not evaluate online assistive control, patients or rehabilitation outcomes.

This research received no funding. I declare no competing interests. I confirm that neither ethics approval nor an exemption was required for this secondary analysis of

publicly available EEG recordings; no new participants were recruited.

I confirm that I am the sole author, have reviewed and approved the final manuscript, and that this work is not under consideration elsewhere. It has not been published in a journal or deposited on a preprint server; a candidate manuscript and reproducibility materials are publicly available on GitHub. AI-assisted work is disclosed in the manuscript Acknowledgements and Methods, including the author-reported use of ChatGPT (GPT-6) and the established OpenAI Codex assistance. No other human assistance is reported.

Thank you for considering the manuscript.

Sincerely,

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